

LUMBERTON, NC, USA

WMO: 723108

Lat: 34.608N

Lon: 79.059W

Elev: 121

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 13776

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	26.0	5.9	7.5	29.6	10.1	9.3	33.9	23.7	48.5	20.6	47.6	4.2	310	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	95.4	76.4	93.0	75.9	90.9	75.3	79.5	89.9	78.4	88.2	77.6	86.9	8.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	140.1	83.5	75.6	134.4	82.1	75.0	131.9	81.5	42.9	89.9	41.9	88.6	40.9	87.0	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	16.9	15.0	15.6	99.3	3.5	2.8	13.1	101.3	11.0	102.9	9.0	104.5	6.4	106.6
			13.9	81.3	3.3	1.3	11.6	82.2	9.6	83.0	7.8	83.8	5.4	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	44.3	47.3	54.1	62.4	70.8	78.7	80.8	79.8	74.2	63.7	53.9	47.5
	DBStd	15.04	10.47	9.12	9.25	7.78	6.41	4.58	3.51	3.86	5.16	8.01	8.60	9.60
	HDD50	689	242	149	66	6	0	0	0	0	0	4	54	168
	HDD65	2679	644	501	353	139	23	0	0	0	5	121	344	549
	CDD50	5510	65	73	194	377	646	861	954	924	725	430	172	89
	CDD65	2022	1	4	16	60	204	411	489	459	279	82	12	5
	CDH74	18579	3	28	139	578	1898	3964	4843	4232	2144	653	79	18
	CDH80	7638	0	1	13	119	689	1811	2217	1867	769	146	5	1
Wind	WSAvg	6.0	6.6	7.0	7.4	7.1	6.2	5.7	5.3	4.7	5.6	5.3	5.4	5.8
Precipitation	PrecAvg	54.0	4.3	3.9	4.5	3.2	4.3	5.6	6.5	6.0	5.0	3.5	3.2	3.4
	PrecMax	69.0	8.6	8.0	11.2	8.0	10.2	14.6	13.1	12.5	16.9	10.6	7.6	8.1
	PrecMin	40.6	0.3	0.9	0.5	0.1	1.0	2.1	1.5	1.1	0.3	0.0	0.8	0.2
	PrecStd	7.4	2.0	1.9	2.6	1.7	2.3	2.5	2.6	2.7	3.4	2.4	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	78.5	81.8	87.6	93.1	98.9	97.8	98.8	93.2	87.8	80.8	76.2
		MCWB	64.4	65.5	64.8	69.6	73.3	75.9	77.3	76.3	73.1	72.9	68.1	67.9
	2%	DB	70.0	72.6	78.3	83.6	90.0	94.9	94.8	94.1	90.4	83.9	76.4	72.3
		MCWB	61.8	61.9	63.9	67.8	71.6	75.5	77.2	76.9	73.5	69.8	65.1	65.3
	5%	DB	65.8	68.0	74.0	80.9	87.5	92.0	92.6	91.4	87.8	81.1	72.9	68.3
		MCWB	59.4	59.1	60.9	66.2	70.7	74.4	76.9	76.6	73.0	68.2	62.8	63.3
	10%	DB	62.1	63.7	70.3	77.4	83.9	89.7	90.3	89.6	84.4	77.5	69.9	64.1
		MCWB	56.1	56.1	59.6	64.3	69.4	73.9	76.3	76.3	72.0	67.5	62.3	59.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	67.5	68.2	73.4	76.0	79.2	81.0	80.9	78.3	76.3	71.2	70.3
		MCDB	70.9	74.1	76.2	81.9	87.3	91.3	92.2	89.9	86.6	80.7	76.4	74.2
	2%	WB	63.9	64.5	66.1	70.8	74.3	77.7	79.7	79.5	76.9	74.3	68.3	66.8
		MCDB	68.3	70.4	73.7	78.9	84.7	89.2	90.5	89.4	84.6	79.3	72.8	70.0
	5%	WB	60.8	61.5	64.2	68.8	72.9	76.8	78.5	78.4	75.6	72.3	66.1	63.8
		MCDB	64.8	66.5	70.7	77.0	82.1	87.3	88.5	87.7	82.7	77.5	70.7	67.5
	10%	WB	57.0	57.5	62.2	66.8	71.4	75.7	77.6	77.5	74.4	70.0	63.5	60.5
		MCDB	61.1	62.9	68.3	74.5	79.8	85.1	87.1	86.3	80.7	75.0	67.8	64.4
Mean Daily Temperature Range	5% DB	MDBR	19.7	21.2	22.5	23.4	22.5	19.8	18.3	18.1	18.8	22.5	23.7	20.1
		MCDBR	23.7	25.8	25.8	25.7	25.4	22.8	20.8	21.1	22.4	24.7	26.1	22.7
		MCWBR	16.6	16.3	13.7	12.3	9.8	7.2	6.8	7.1	8.4	11.7	15.3	16.3
	5% WB	MCDBR	20.6	23.8	21.2	21.1	20.3	19.3	18.7	18.2	16.8	17.0	19.5	20.4
MCWBR		16.3	17.5	13.7	11.5	8.6	6.9	7.1	7.0	7.4	9.3	13.9	16.4	
Clear-Sky Solar Irradiance	taub	0.321	0.330	0.368	0.404	0.466	0.495	0.541	0.515	0.431	0.395	0.344	0.319	
	taud	2.490	2.459	2.381	2.276	2.153	2.113	2.004	2.067	2.299	2.364	2.466	2.531	
	Ebn at Noon	281	291	287	280	263	254	241	246	263	264	270	275	
	Edh at Noon	27	31	36	42	48	50	56	51	39	34	27	24	
All-Sky Solar Radiation	RadAvg	859	1066	1411	1767	1927	2016	1924	1732	1478	1224	943	752	
	RadStd	49	110	109	117	152	131	121	87	121	147	76	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (64 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+216

Nomenclature: See separate page